

1.2 COURSE PACING GUIDES

IST Semester 1

Georgia Standards

- IT-IST-1 through IT-IST-7

Unit / Standards	Timeline
Orientation & Career Exploration (IT-IST-1, IT-IST-7)	Week 1
Digital Citizenship & Safety (IT-IST-2)	Weeks 2-3
Technology Systems & Troubleshooting (IT-IST-3)	Weeks 4-5
Programming Foundations with Micro:bit (IT-IST-4)	Weeks 6-8
Web Development & UI Design (IT-IST-5)	Weeks 9-10
Robotics & Automation with Finch (IT-IST-3, IT-IST-4)	Weeks 11-12
Career Exploration & Capstone Project (IT-IST-1 through IT-IST-7)	Weeks 13-18

Major Projects

- Technology Career Research
- Micro:bit Programming Project
- Personal Website Development

FBLA Connections

1. Introduction to Information Technology
2. Website Coding & Development
3. Coding & Programming

CSP Semester 2

Georgia Standards

- IT-CSP-1 through IT-CSP-9

Weeks	Unit / Standards
1-4	IT-IST-4 Advanced Programming Applications
5-7	IT-IST-5 Advanced Web Design & User Experience
8-10	IT-IST-6 Data Analysis & Technology Research
11-13	IT-IST-7 Collaborative Technology Project
14-18	Comprehensive Capstone Project (IT-IST 1-7)

Major Projects

- JavaScript Application
- Data Visualization Project
- Cybersecurity Investigation
- Computing Ethics Analysis

FBLA Connections

1. Computer Problem Solving
2. Coding & Programming

AP Computer Science Principles

Georgia Standards

- AP CSP Big Idea 1: Creative Development
- AP CSP Big Idea 2: Data
- AP CSP Big Idea 3: Algorithms and Programming
- AP CSP Big Idea 4: Computer Systems and Networks
- AP CSP Big Idea 5: Impact of Computing

Semester 1

Weeks	Unit / Standards	Major Project / FBLA Connection
1-2	Computing Innovations & Collaboration (Big Ideas 1,5)	Explore FBLA Computer Problem Solving, Coding & Programming, and Intro to Programming events
3-6	Python Fundamentals (Big Idea 3)	Python Mini Projects and Computational Thinking Challenges
7-9	Algorithms & Problem Solving (Big Idea 3)	Powerball vs Bitcoin Mining Investigation
10-12	Data & Information (Big Idea 2)	Data Analysis Project using Python
13-15	Networks & Cybersecurity (Big Idea 4)	Cybersecurity Awareness Activity and FBLA Cyber Security Event Exploration
16-18	Impacts of Computing (Big Idea 5)	Computational Innovation Research Presentation

Semester 2:

Weeks	Unit / Standards	Major Project / FBLA Connection
1-3	Functions and Abstraction (Big Idea 3)	Python Application Development

4-6	Advanced Programming (Big Ideas 1,3)	Software Development Team Project
7-9	Data Analysis & Innovation (Big Ideas 2,5)	Computational Innovation Research
10-14	Create Performance Task (Big Ideas 1-5)	College Board Create Task
15-17	AP Review	AP Classroom MCQs/FRQs
18	Innovation & Career Exploration	Finch Robotics, Industry Guest Speakers, FBLA Preparation

Major Projects

- Roblox Interactive Experience
- Data Science Investigation
- Computational Artifact Development
- AP Create Performance Task
- AP Exam Preparation Portfolio

FBLA Connections

1. Computer Problem Solving
2. Coding & Programming
3. Cyber Security

AP Computer Science A

Georgia Standards

- AP Computer Science A Framework

Semester 1

Weeks	Unit / Standards
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1-4	Unit 1: Using Objects and Methods (AP CSA Unit 1: Primitive Types, Objects, Method Calls)
5-8	Unit 2: Selection and Iteration (AP CSA Unit 2: Using Objects and Unit 3: Boolean Expressions and if Statements)
9-12	Unit 3: Class Creation (AP CSA Unit 5: Writing Classes)
13-16	Unit 4: Real-World Programming & APIs (Application of Units 1-5 through Java programming, data retrieval, JSON interpretation, debugging, testing, and software development practices)
17-18	Semester 1 Review and Assessment (Units 1-5)

Semester 2

Weeks	Unit / Standards
1-3	Unit 5: Data Collections (AP CSA Unit 6: Array and Unit 7: ArrayList)
4-6	Unit 6: Arrays (AP CSA Unit 6: Array)
7-8	Unit 7: ArrayList (AP CSA Unit 7: ArrayList)
9-10	Unit 8: 2D Arrays (AP CSA Unit 8: 2D Array)
11-13	Unit 9: Inheritance (AP CSA Unit 9: Inheritance)
14-15	Unit 10: Recursion (AP CSA Unit 10: Recursion)
16-17	AP Exam Review (Comprehensive Review of Units 1-10)
18	Post-AP Computer Science Applications (Integration and Extension of AP CSA Concepts)

Major Projects

- Java Fundamentals Application
- Selection and Iteration Project
- Object-Oriented Programming Project
- Data Collections Application

- AP Practice Portfolio

FBLA Connections

1. Coding & Programming
2. Computer Problem Solving
3. Programming Concepts

Game Design and Simulation

Georgia Standards

- GDAS1 Employability Skills
- GDAS2 Introduction to the Game Industry
- GDAS3 Game Design Process
- GDAS4 Game Asset Development
- GDAS5 Programming and Game Development
- GDAS6 Testing, Debugging, and Deployment
- GDAS7 Career Development and Industry Certification

Semester 1

Weeks	Unit / Standards
1-2	GDAS1 Career Exploration & Industry Overview
3-4	GDAS2 History of Games and Game Analysis
5-7	GDAS3 Game Design Documentation & Planning
8-10	GDAS5 Unity Development Fundamentals
11-13	GDAS4 Visual Design, Art, Audio, and User Experience
14-18	Unity Prototype Development Project (GDAS3, GDAS4, GDAS5)

Major Projects

- Unity Environment Project
- Interactive Prototype
- Gameplay Systems Project
- User Interface Design Project
- Optimization Analysis
- Original Unity Game

Semester 2

Weeks	Unit / Standards
1-3	GDAS5 Intermediate Unity Development
4-6	GDAS4 Asset Integration & User Interface Design
7-9	GDAS3 Gameplay Systems & Game Mechanics
10-12	GDAS6 Testing, Debugging, and Optimization
13-15	GDAS7 Professional Practices & Publishing
16-18	Final Game Development Capstone (GDAS1-GDAS7)

Major Projects

- Digital Dopamine Game Design Document (GDD)
- Asset Library Development
- Unity Certification Practice Activities
- User Testing Report
- Final Unity Game Project
- End-of-Course Project Presentation

FBLA Connections

1. Game Design

2. Computer Game & Simulation Programming